

Term 3 Practice Activity: C6 Standardisation in Context

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Evaluated criteria

- C6. Uses normal distributions in contextual situations.

Contents

Evaluated criteria

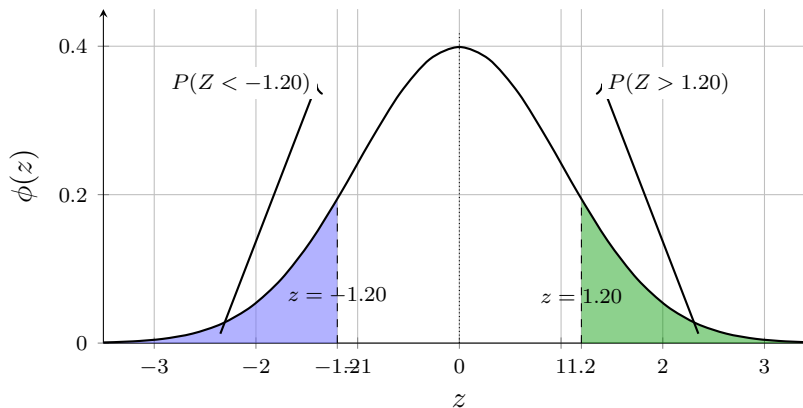
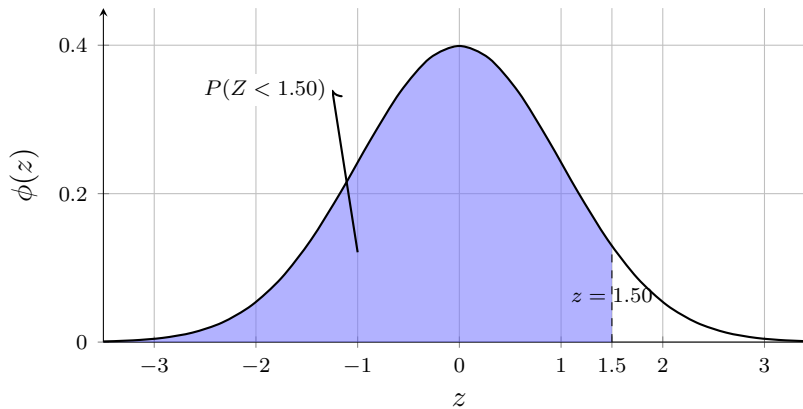
- C6.

Standard normal table (values of $\Phi(z)$).

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545

1 Problem 1: Normal Distribution Left Tail with Positive and Negative z-values

Problem. Exam scores follow $X \sim N(70, 8^2)$. Find $P(X < 82)$ and $P(X < 60.4)$.



$$Z = \frac{X - 70}{8}.$$

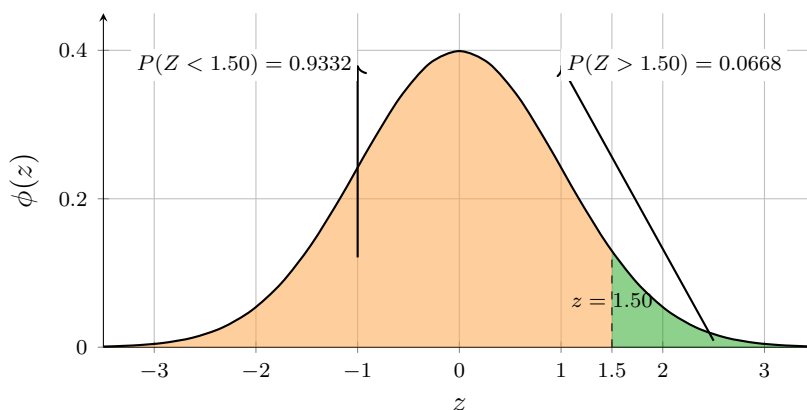
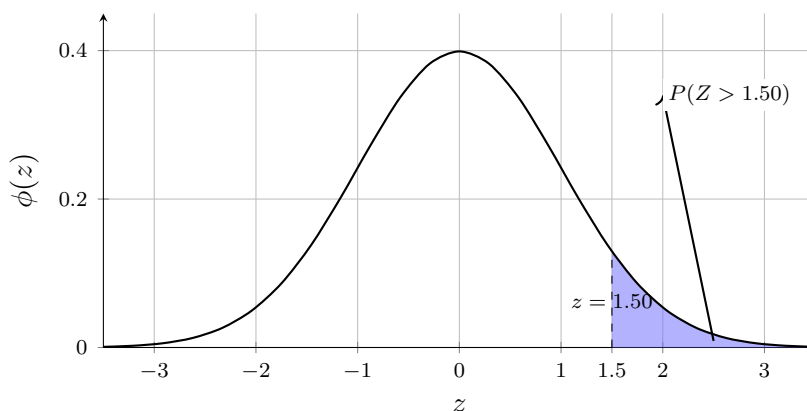
$$P(X < 82) = P\left(Z < \frac{82 - 70}{8}\right) = P(Z < 1.50) = 0.9332.$$

$$P(X < 60.4) = P\left(Z < \frac{60.4 - 70}{8}\right) = P(Z < -1.20) = 1 - 0.8849 = 0.1151.$$

Interpretation. About 93.32% score below 82, while 11.51% score below 60.4.

2 Problem 2: Normal Distribution Right Tail with Positive and Negative z-values

Problem. Delivery times follow $X \sim N(56, 5^2)$ minutes. Find $P(X > 63.5)$ and $P(X > 52)$.



$$Z = \frac{X - 56}{5}.$$

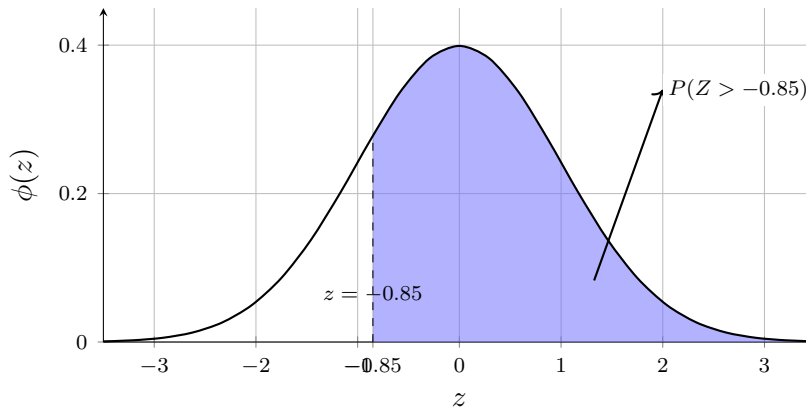
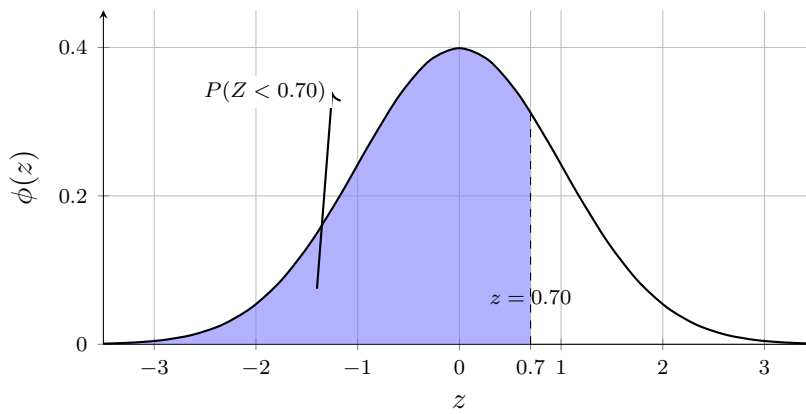
$$P(X > 63.5) = P\left(Z > \frac{63.5 - 56}{5}\right) = P(Z > 1.50) = 1 - 0.9332 = 0.0668.$$

$$P(X > 52) = P\left(Z > \frac{52 - 56}{5}\right) = P(Z > -0.80) = \Phi(0.80) = 0.7881.$$

Interpretation. 6.68% take more than 63.5 min, and 78.81% take more than 52 min.

3 Problem 3: Normal Distribution Mixed Left-Tail and Right-Tail Reasoning

Problem. Production weights follow $X \sim N(500, 20^2)$ g. Find $P(X < 514)$ and $P(X > 483)$.



$$Z = \frac{X - 500}{20}.$$

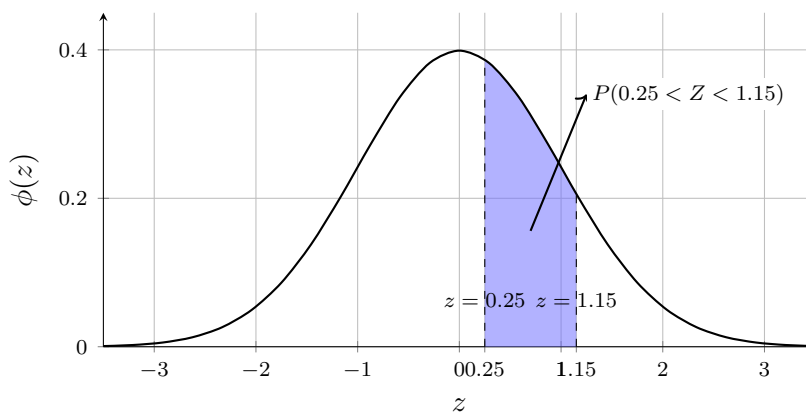
$$P(X < 514) = P\left(Z < \frac{514 - 500}{20}\right) = P(Z < 0.70) = 0.7580.$$

$$P(X > 483) = P\left(Z > \frac{483 - 500}{20}\right) = P(Z > -0.85) = \Phi(0.85) = 0.8023.$$

Interpretation. 75.80% are below 514 g, and 80.23% are above 483 g.

4 Problem 4: Normal Distribution Central Band with Positive z-bounds

Problem. Weekly app usage follows $X \sim N(14, 4^2)$ hours. Find $P(15 < X < 18.6)$.



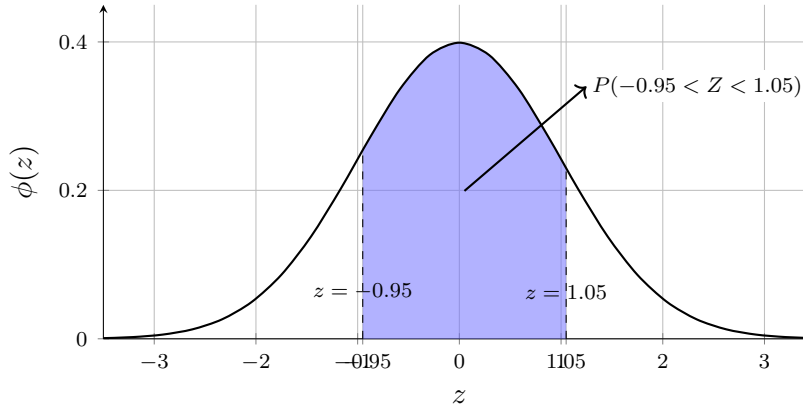
$$Z = \frac{X - 14}{4}, \quad P(15 < X < 18.6) = P(0.25 < Z < 1.15).$$

$$P = \Phi(1.15) - \Phi(0.25) = 0.8749 - 0.5987 = 0.2762.$$

Interpretation. 27.62% of students use the app between 15 h and 18.6 h weekly.

5 Problem 5: Normal Distribution Central Band with One Negative and One Positive z-bound

Problem. Commute times follow $X \sim N(40, 10^2)$ minutes. Find $P(30.5 < X < 50.5)$.



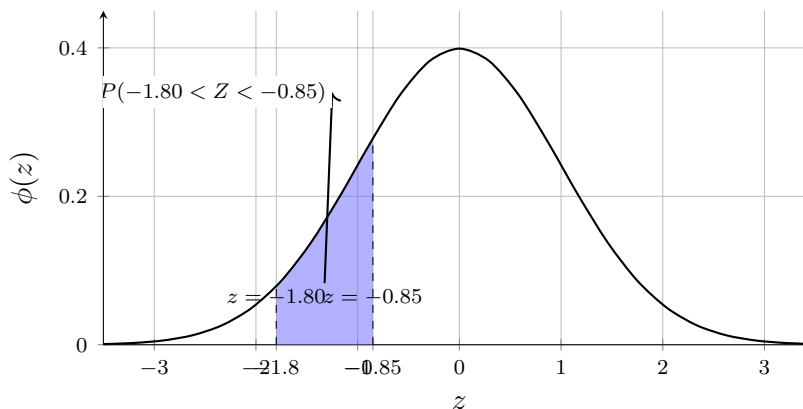
$$Z = \frac{X - 40}{10}, \quad P(30.5 < X < 50.5) = P(-0.95 < Z < 1.05).$$

$$P = \Phi(1.05) - \Phi(-0.95) = 0.8531 - (1 - 0.8289) = 0.6820.$$

Interpretation. 68.20% of commutes are between 30.5 and 50.5 minutes.

6 Problem 6: Normal Distribution Central Band with Both Negative z-bounds

Problem. Monthly sales follow $X \sim N(120, 20^2)$ units. Find $P(84 < X < 103)$.



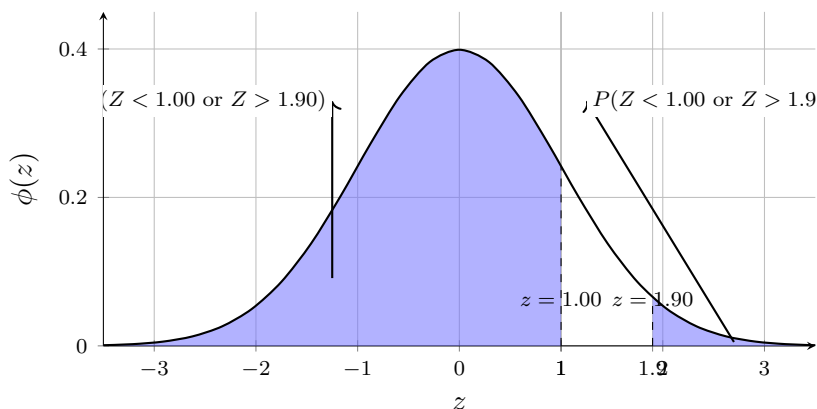
$$Z = \frac{X - 120}{20}, \quad P(84 < X < 103) = P(-1.80 < Z < -0.85).$$

$$P = \Phi(-0.85) - \Phi(-1.80) = (1 - 0.8023) - (1 - 0.9641) = 0.1618.$$

Interpretation. 16.18% of months have sales between 84 and 103 units.

7 Problem 7: Normal Distribution Two Regions with Positive z-bounds

Problem. Household electricity use follows $X \sim N(400, 50^2)$ kWh. Find $P(X < 450 \text{ or } X > 495)$.



$$Z = \frac{X - 400}{50}, \quad P(X < 450 \text{ or } X > 495) = P(Z < 1.00 \text{ or } Z > 1.90).$$

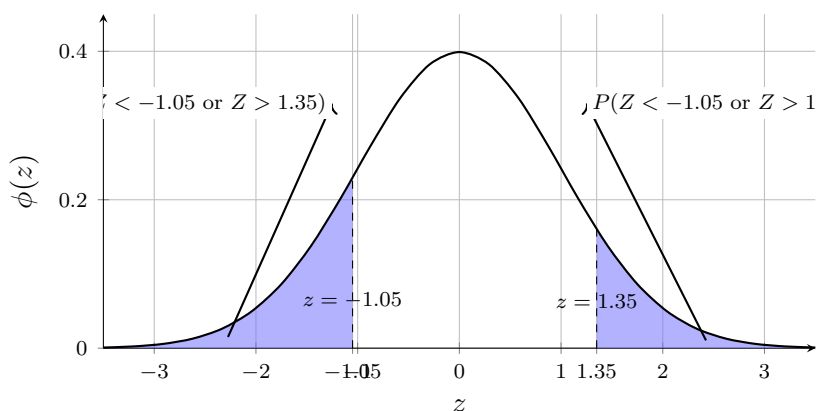
The regions are disjoint, so

$$P = \Phi(1.00) + (1 - \Phi(1.90)) = 0.8413 + 0.0287 = 0.8700.$$

Interpretation. 87.00% of homes are below 450 kWh or above 495 kWh.

8 Problem 8: Normal Distribution Two Tails with One Negative and One Positive z-bound

Problem. Scholarship scores follow $X \sim N(300, 20^2)$. Find $P(X < 279 \text{ or } X > 327)$.



$$Z = \frac{X - 300}{20}, \quad P(X < 279 \text{ or } X > 327) = P(Z < -1.05 \text{ or } Z > 1.35).$$

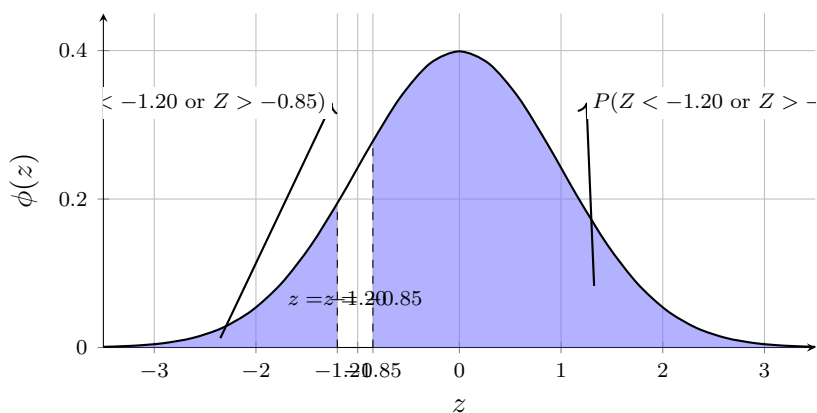
The regions are disjoint, so

$$P = \Phi(-1.05) + (1 - \Phi(1.35)) = (1 - 0.8531) + (1 - 0.9115) = 0.2354.$$

Interpretation. 23.54% are either below 279 or above 327 points.

9 Problem 9: Normal Distribution Two Regions with Both Negative z-bounds

Problem. Customer waiting times follow $X \sim N(90, 20^2)$ seconds. Find $P(X < 66 \text{ or } X > 73)$.



$$Z = \frac{X - 90}{20}, \quad P(X < 66 \text{ or } X > 73) = P(Z < -1.20 \text{ or } Z > -0.85).$$

The regions are disjoint, so

$$P = \Phi(-1.20) + (1 - \Phi(-0.85)) = 0.1151 + 0.8023 = 0.9174.$$

Interpretation. 91.74% wait below 66 s or above 73 s.